

# TAK Server 'X-Forwarded-For' Logging Enablement

TAK Server's HTTP access log records only the address of the device that connects to it. Behind a load balancer or reverse proxy such as a BIG-IP, every entry shows the proxy's address, even when the proxy sends the client address in the X-Forwarded-For header. The log format is set in compiled Java code and cannot be changed through CoreConfig.xml or the startup scripts.

This procedure adds the X-Forwarded-For value to each log entry by patching the installed takserver.war. It extracts the one class that defines the tomcat log format, modifies the default log pattern using a Python script, and places the class back into the compiled war. The procedure has been validated against TAK Server [5.5-RELEASE-82](#) and should work for subsequent releases, provided that the TAK developers do not change the Tomcat architecture.

## Patching the takserver.war using jar and python

### Non-TAK-prod RHEL server tools preparation *(used for patching the war)*

1. Install the required packages on the RHEL server or equivalent

```
dnf install -y java-21-openjdk-devel python3 unzip binutils
```

2. Create a working directory

```
mkdir -p /var/tmp/takserver-xff
```

### Record and back up the original files

3. On the TAK server, save the md5 hashes of the original files

```
cd /opt/tak
md5sum takserver.war CoreConfig.xml > takserver-original.md5
cat takserver-original.md5
```

4. On the TAK server, back up the original files

```
cp -a /opt/tak/takserver.war /opt/tak/takserver.war.backup
cp -a /opt/tak/CoreConfig.xml /opt/tak/CoreConfig.xml.backup
```

5. On the TAK server, verify the backups against the saved hashes – Both lines must report OK

```
cd /opt/tak
sed 's/$/.backup/' takserver-original.md5 | md5sum -c -
```

### Copy the war to the RHEL server

6. From the TAK server, copy the war and the hash file to the Tak build server or a non-Tak RHEL server

```
cd /opt/tak
scp takserver.war takserver-original.md5 root@<rhel-server>:/var/tmp/takserver-xff/
```

7. On the Tak build server or a non-Tak RHEL server, verify the copied war is the unmodified original – The line must report OK

```
cd /var/tmp/takserver-xff
grep 'takserver.war$' takserver-original.md5 | md5sum -c -
```

### Patch the access log pattern in the war

8. Extract the class file that defines the Tomcat log format from the war

jar recreates the WEB-INF/classes/tak/server path on disk, which is the path it needs to place the class back into the war

```
cd /var/tmp/takserver-xff
jar xf takserver.war WEB-INF/classes/tak/server/CustomizeEmbeddedTomcatContainer.class
```

9. Add the X-Forwarded-For field to the log pattern in the class file – **Run once only**

The class file is binary and must not be edited in vi or another text editor. The script inserts the field, corrects the stored length of the pattern string, and stops without writing if the stock pattern is not found exactly once.

```
python3 - <<'EOF'
import struct
p = 'WEB-INF/classes/tak/server/CustomizeEmbeddedTomcatContainer.class'
d = open(p, 'rb').read()
old = b'remoteip=%a user='
new = b'remoteip=%a xff=%{X-Forwarded-For}i user='
assert d.count(old) == 1
i = d.index(old)
s = d.rfind(b'request: method=', 0, i)
assert d[s-3] == 1
n = struct.unpack('>H', d[s-2:s])[0]
d = d[:s-2] + struct.pack('>H', n + len(new) - len(old)) + d[s:i] + new + d[i+len(old):]
open(p, 'wb').write(d)
EOF
```

10. Place the modified class file back into the war and remove the extracted copy

```
jar uf takserver.war WEB-INF/classes/tak/server/CustomizeEmbeddedTomcatContainer.class
rm -rf WEB-INF
```

11. Verify the patch – The output must be 1 before proceeding further

```
unzip -p takserver.war WEB-INF/classes/tak/server/CustomizeEmbeddedTomcatContainer.class | strings | grep -c 'xff='
```

- 12.** Save the md5 hash of the patched war

```
cd /var/tmp/takserver-xff  
md5sum takserver.war > takserver-xff.md5  
cat takserver-xff.md5
```

### Transfer the patched war to the TAK server

- 13.** On the TAK server, create a staging directory

```
mkdir -p /var/tmp/takserver-xff
```

- 14.** From the Tak build server or a non-Tak RHEL server, copy the patched war and its hash file to the TAK server

```
cd /var/tmp/takserver-xff  
scp takserver.war takserver-xff.md5 root@<tak-server>:/var/tmp/takserver-xff/
```

- 15.** On the TAK server, verify the transfer – The line must report OK

```
cd /var/tmp/takserver-xff  
md5sum -c takserver-xff.md5
```

### Install the patched war

- 16.** On the TAK server, stop the service

```
systemctl stop takserver
```

- 17.** Install the patched war and set ownership

```
cp /var/tmp/takserver-xff/takserver.war /opt/tak/takserver.war  
chown tak:tak /opt/tak/takserver.war
```

- 18.** Verify the installed war – The line must report OK

```
cd /opt/tak  
md5sum -c /var/tmp/takserver-xff/takserver-xff.md5
```

- 19.** Check whether the HTTP access log is enabled

```
grep -n '<logging' /opt/tak/CoreConfig.xml
```

- 20.** If the previous step printed nothing, add the logging element

If it printed a logging line without `httpAccessEnabled="true"`, add that attribute to the existing line by hand and skip this command

```
sed -i 's#</Configuration>#    <logging httpAccessEnabled="true"/>\n</Configuration>#'
/opt/tak/CoreConfig.xml
```

- 21.** Start the service

```
systemctl start takserver
```

## Logging Verification

- 22.** Confirm the API process reports the access log as enabled

```
grep 'http access logging' /opt/tak/logs/takserver-api.log | tail -1
```

Expected output:

```
... http access logging enabled: true
```

- 23.** Review the access log

```
tail -f /opt/tak/logs/takserver-api-access.log
```

Expected entry:

```
... client: remoteip=<BIG-IP address> xff=<client address> user=... useragent="..."
```

## Rollback

- 24.** On the TAK server, restore the original files and restart the service

```
systemctl stop takserver
cp -a /opt/tak/takserver.war.backup /opt/tak/takserver.war
cp -a /opt/tak/CoreConfig.xml.backup /opt/tak/CoreConfig.xml
systemctl start takserver
```

- 25.** Verify the restored files against the original hashes – Both lines must report OK

```
cd /opt/tak
md5sum -c takserver-original.md5
```

## Appendix: References

**TAK Server source repository** – public source for TAK Server

<https://github.com/TAK-Product-Center/Server>

**TAK Server 5.5-RELEASE-82** – the release this procedure was validated against

<https://github.com/TAK-Product-Center/Server/releases/tag/5.5-RELEASE-82>

**CustomizeEmbeddedTomcatContainer.java** – the class that sets the access log pattern

<https://github.com/TAK-Product-Center/Server/blob/5.5-RELEASE-82/src/takserver-core/src/main/java/tak/server/CustomizeEmbeddedTomcatContainer.java>

**CoreConfig.xsd** – defines the logging element and its httpAccessEnabled attribute

<https://github.com/TAK-Product-Center/Server/blob/5.5-RELEASE-82/src/takserver-common/src/main/xsd/CoreConfig.xsd>

**Apache Tomcat Access Log Valve** – pattern codes, including %a and %{xxx}i for request headers

[https://tomcat.apache.org/tomcat-10.1-doc/config/valve.html#Access\\_Log\\_Valve](https://tomcat.apache.org/tomcat-10.1-doc/config/valve.html#Access_Log_Valve)

**X-Forwarded-For header** – description of the header and its format

<https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/X-Forwarded-For>

**TAK.gov** – source of the pre-built TAK Server packages

<https://tak.gov>